

SESSION 3

RAD-541 Magnetic Resonance Imaging II

Frequency Encoding and Phase Encoding

How MRI assigns in-plane position using frequency during readout and phase before readout

Course	Magnetic Resonance Imaging II
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Suggested study time	240–300 minutes

Learning Objectives

By the end of this session, you should be able to:

1. explain why slice selection alone is insufficient to form a 2D MR image;
2. describe how the frequency-encoding or readout gradient makes frequency depend on in-plane position;
3. explain what receiver bandwidth means and distinguish it from RF transmit bandwidth;
4. describe how a temporary phase-encoding gradient creates position-dependent phase shifts;
5. explain why phase encoding must be repeated with different gradient amplitudes;
6. compare frequency encoding and phase encoding in terms of timing, information and image consequences;
7. relate frequency samples and phase-encoding steps to matrix and spatial resolution;
8. explain why the radiographer may change the phase and frequency directions; and
9. predict common image effects associated with phase direction, frequency direction and receiver bandwidth.

The central idea

After a slice has been selected, MRI still has to identify where signal came from *within* that slice. One in-plane direction is encoded by making spins at different positions precess at different **frequencies during signal readout**. The perpendicular in-plane direction is encoded by giving spins at different positions different **phase angles before readout**. Each voxel is therefore identified by a unique combination of frequency and phase information.

1 Session Orientation

Session 2 answered the first spatial question: *which slice should be excited?* Session 3 answers the two questions that remain inside that selected slice: *where is the signal along one in-plane axis, and where is it along the perpendicular axis?*

A useful map for the whole session is:

selected slice → frequency/readout encoding
phase encoding → raw spatial-frequency data → image reconstruction.

Frequency encoding creates a position-dependent frequency pattern while the signal is sampled. Phase encoding briefly creates position-dependent frequency differences *before* readout; those differences accumulate into phase offsets that remain as spatial labels when the gradient is removed. Keeping this timing distinction clear prevents many later MRI misconceptions.

Why this matters at the console

The radiographer routinely changes phase direction, phase FOV, matrix and receiver bandwidth. These are not isolated menu options: they alter how spatial information is encoded, how long the acquisition takes, and where artifacts such as wrap, motion ghosting and chemical shift appear.

2 From slice selection to in-plane localization

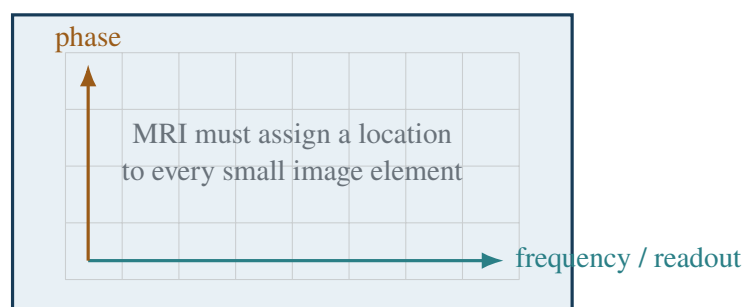
Session 2 explained how a slice-select gradient and an RF pulse select a thin section of anatomy. Imagine, however, that the selected axial slice contains the whole brain. The receiver coil detects a mixture of MR signals from many thousands of locations within that slice. Slice selection tells us that the signal came from the selected level, but it does not yet tell us whether a particular component came from the patient's left or right, anterior or posterior.

MRI therefore performs two additional spatial-encoding operations. In a conventional 2D Cartesian acquisition:

- one in-plane axis is assigned to **frequency encoding**, also called **readout encoding**; and
- the perpendicular in-plane axis is assigned to **phase encoding**.

The roles are functional rather than permanently tied to scanner *x*, *y* or *z*. If the radiographer swaps phase and frequency directions, the scanner changes which physical gradient combination performs each task.

One already-selected 2D slice



Three dimensions, three encoding jobs

In basic 2D MRI, spatial localization can be remembered as: **slice selection chooses the section; frequency encoding identifies one in-plane coordinate; phase encoding identifies the other in-plane coordinate.**

3 Frequency encoding: turning position into frequency

Frequency encoding uses the same physical principle introduced during slice selection: a magnetic-field gradient makes Larmor frequency vary with position. The difference is *when* and *why* the gradient is used.

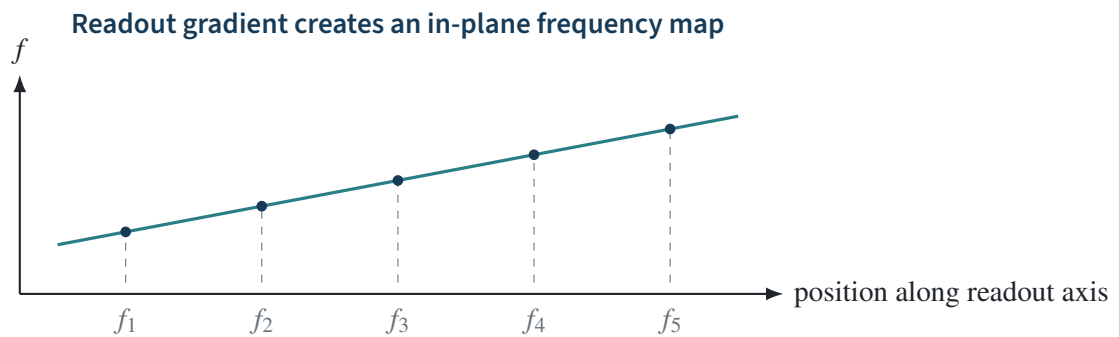
During signal acquisition, the scanner applies a gradient along the chosen frequency direction. This gradient is usually called the **readout gradient**, G_R . If the readout direction is represented by x , then

$$B(x) = B_0 + G_R x$$

and therefore

$$f(x) = \gamma (B_0 + G_R x) = f_0 + \gamma G_R x.$$

The key consequence is simple: **different positions along the readout direction now produce different frequencies**. The received MR signal contains all these frequencies mixed together. The scanner does not listen to one location at a time. Instead, the sampled echo is a composite waveform containing contributions from many positions simultaneously. Fourier analysis separates the frequency components of that waveform, allowing frequency to be translated back into position along the readout axis.



Each position along the gradient axis is assigned a different range of resonance frequencies.

3.1 Why it is called readout encoding

The readout gradient is active while the MR echo is being sampled by the receiver. This is why frequency encoding is also called **readout encoding**. The scanner records a sequence of signal samples while the gradient is on. Those samples contain information from all positions along the readout axis at the same time.

For a simple acquisition, the number of digital samples collected across one echo commonly corresponds to the frequency-encoding matrix dimension. For example, a matrix written as 256×256 may contain 256 sampled positions in the frequency direction and 256 encoded positions in the phase direction, although manufacturer display conventions vary.

Frequency encoding does not mean “one voxel emits only one frequency”

A voxel has finite size and therefore contains a small range of frequencies when a gradient is present. Frequency encoding should be understood as assigning *position-dependent frequency bands* along one axis, not as giving every proton a perfectly unique single frequency.

4 Why frequency encoding alone is not enough

Suppose the readout gradient runs from left to right. Every location in the same vertical column experiences approximately the same readout frequency even though some signal arises anteriorly and some posteriorly. Frequency therefore identifies the *column*, but a second independent label is needed to distinguish the rows.

Two locations with the same readout frequency

Imagine two voxels that lie at the same left–right coordinate but at different anterior–posterior positions. During a left–right readout gradient, both receive the same frequency label. Their signals cannot be separated by frequency alone. Phase encoding supplies the second coordinate needed to distinguish them.

5 What does phase mean?

Phase describes the angular position of a rotating transverse magnetization vector within its cycle. Two spins can precess at the same frequency yet point in different directions at a particular instant; they then have the same frequency but different phase.

A clock-face analogy is useful. Two clocks can tick at exactly the same rate but show different hand positions. **Frequency** is comparable to how fast the hand rotates; **phase** is comparable to where the hand points at one instant.

Frequency and phase are independent labels

Frequency tells us the rate of rotation. Phase tells us the angular position within that rotation. MRI exploits both quantities because one frequency label alone cannot uniquely identify a point in a two-dimensional slice.

6 Phase encoding: creating a position-dependent phase shift

Phase encoding uses a gradient differently from frequency encoding. A **phase-encoding gradient**, G_P , is switched on for a short period before signal readout. While it is on, spins at different positions along the phase direction experience slightly different magnetic fields and therefore precess at slightly different frequencies.

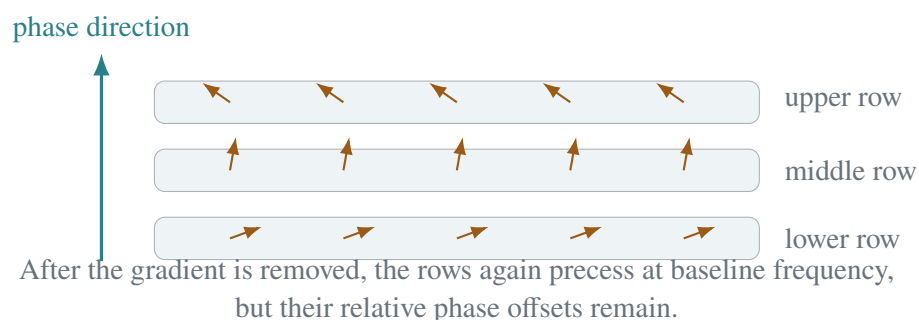
Because they temporarily precess at different rates, they accumulate different phase angles. When the phase-encoding gradient is switched off, the spins return to their baseline frequencies, but the **phase differences remain**. In this way, position has been stored as phase information.

A useful conceptual relationship is

$$\Delta\phi(y) = \gamma G_P y \tau,$$

where y is position along the phase direction and τ is the duration of the phase-encoding gradient. The quantity $G_P \tau$ is the **gradient area**. For a given position, the accumulated phase therefore depends on the area under the phase-encoding gradient, not only on its peak amplitude. A stronger gradient applied briefly can produce the same ideal phase shift as a weaker gradient applied for longer if the gradient area is the same.

A temporary phase gradient leaves a lasting phase pattern



Frequency is present during readout; phase is prepared before readout

For conventional Cartesian imaging, the frequency/readout gradient is applied while the signal is sampled. The phase-encoding gradient is usually applied before readout and then switched off. Its spatial information survives as a phase difference.

7 Why phase encoding must be repeated

One phase-encoding gradient creates one pattern of phase offsets across the image, but the receiver still records the combined signal from all rows. To resolve all rows mathematically, the sequence is repeated with a series of different phase-encoding gradient areas: negative, near zero and positive values.

Each repetition therefore provides a different spatial encoding condition. In basic Cartesian MRI, a different phase-encoding value selects a different line of raw data in k-space. The full image requires enough independent phase-encoding measurements to describe the prescribed phase matrix.

Phase-encoding condition	What changes
Negative gradient area	Spins accumulate one direction of position-dependent phase shift
Near-zero gradient area	Minimal additional phase shift from the phase-encoding lobe
Positive gradient area	Spins accumulate the opposite direction of position-dependent phase shift

7.1 Why phase encoding affects scan time

In a simple 2D sequence, each new phase-encoding step generally requires another sequence repetition. Therefore, increasing the number of phase-encoding steps usually increases acquisition time. A simplified relationship often used for teaching is

$$\text{scan time} \approx TR \times N_{PE} \times NEX,$$

where N_{PE} is the number of phase-encoding steps and NEX/NSA is the number of signal averages. Real sequences may modify this relationship. For example, fast/turbo spin-echo sequences can acquire several phase-encoded k-space lines within one TR using an echo train, while parallel imaging and partial Fourier can reduce the number of measurements required. The core principle nevertheless remains: **the number of phase-encoding measurements is a major determinant of scan time.**

Radiographer's parameter choices

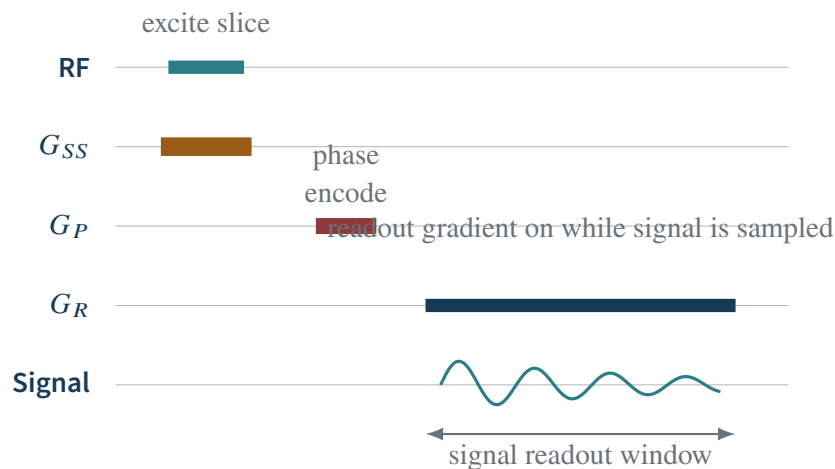
If you increase the phase matrix from 192 to 256 while keeping the rest of a basic sequence unchanged, you usually request more phase-encoding steps. This can improve in-plane resolution in the phase direction, but it also tends to increase scan time unless another parameter or acceleration method compensates.

Matrix and scan-time reasoning

If a basic sequence uses $TR = 500$ ms, one signal average and 192 phase-encoding steps, the simplified scan time is about $0.5 \times 192 = 96$ s. Increasing only the phase matrix to 256 gives about 128 s. This simplified example shows why phase resolution and scan time are closely linked.

8 A complete 2D encoding sequence

It is useful to see the three spatial operations in time rather than as isolated definitions. A simplified sequence may be represented as follows.



The exact gradient lobes and timing depend on the pulse sequence, but the conceptual order is robust: **select a region, impose phase information, then read the signal while frequency encoding is active.**

9 Frequency encoding versus phase encoding

Feature	Frequency / readout encoding	Phase encoding
Main spatial label	Resonance frequency	Phase angle
Gradient timing	Active during signal readout	Briefly applied before readout
How position is encoded	Position changes frequency continuously along the readout axis	Position accumulates a different phase shift
Repeated amplitudes?	One readout gradient waveform samples many positions during one echo	Usually many different amplitudes across repetitions
Matrix link	Number of readout samples	Number of phase-encoding steps
Strong scan-time effect	Usually smaller direct effect	Major effect in conventional 2D imaging
Common artifact direction	Chemical shift / readout distortion	Motion ghosting and phase wrap often appear here

Phase is not image brightness

The word “phase” here refers to the angular relationship of transverse magnetization, not to a bright or dark image phase. Phase encoding is a spatial-labeling process that occurs before image reconstruction.

10 Receiver bandwidth: the bandwidth of readout

Session 2 used **RF transmit bandwidth** to explain slice thickness. Frequency encoding introduces a different parameter: **receiver bandwidth**.

Receiver bandwidth describes the range of signal frequencies accepted and sampled during readout. Because the readout gradient deliberately spreads resonance frequencies across the field of view, the receiver must cover that spatial frequency range. Receiver bandwidth is linked to the sampling interval, or **dwelt time**: a wider bandwidth requires faster sampling with a shorter dwell time, whereas a narrower bandwidth allows slower sampling.

For a simple linear readout gradient, the frequency span across a field of view can be estimated from

$$\Delta f_{FOV} = \gamma G_R FOV_R.$$

Thus, a larger readout gradient or larger readout FOV creates a wider spread of frequencies.

Frequency spread across the readout FOV

Suppose $G_R = 4 \text{ mT/m} = 0.004 \text{ T/m}$ and the readout FOV is $24 \text{ cm} = 0.24 \text{ m}$. For hydrogen,

$$\Delta f_{FOV} = (42.58 \times 10^6)(0.004)(0.24) \approx 40.9 \text{ kHz}.$$

The receiver therefore needs to handle a frequency range on the order of tens of kilohertz for this readout example.

10.1 Total bandwidth and bandwidth per pixel

If a total receiver bandwidth is distributed across N_R frequency-encoding samples, the approximate bandwidth per pixel is

$$BW_{pixel} \approx \frac{BW_{total}}{N_R}.$$

For example, a total receiver bandwidth of 32 kHz with 256 readout samples corresponds to about 125 Hz/pixel.

Different manufacturers display receiver bandwidth in different forms, so radiographers should always check whether the console is showing total bandwidth, Hz/pixel, or another related specification.

Bandwidth per pixel also gives a useful way to think about chemical shift. If fat and water differ by a particular number of hertz, the approximate displacement in pixels along the readout direction is

$$\text{chemical-shift displacement (pixels)} \approx \frac{\Delta f_{fat-water}}{BW_{pixel}}.$$

A higher Hz/pixel therefore reduces the number of pixels of chemical-shift displacement.

10.2 What happens when receiver bandwidth is changed?

Receiver bandwidth is an important image-quality trade-off.

- **Higher receiver bandwidth** uses a shorter dwell time and accepts a wider range of noise frequencies. It tends to reduce frequency-direction chemical shift and some susceptibility/readout distortion, but generally lowers SNR when other factors are unchanged.
- **Lower receiver bandwidth** uses a longer dwell time and accepts less noise bandwidth, so SNR generally improves when other factors are unchanged. The trade-off is greater frequency-related displacement/distortion and potentially longer minimum readout timing.

Two bandwidths, two different jobs

Transmit bandwidth belongs mainly to RF excitation and slice selection. **Receiver bandwidth** belongs to signal acquisition and frequency encoding. Both are measured in frequency units, but they describe different parts of the sequence.

11 FOV, matrix and pixel size

After slice thickness has defined the through-plane dimension, in-plane pixel dimensions depend on FOV and matrix.

For a rectangular image,

$$\text{pixel size}_R = \frac{FOV_R}{N_R}, \quad \text{pixel size}_P = \frac{FOV_P}{N_P}.$$

For example, a 256×256 matrix over a 256 mm square FOV gives an in-plane pixel dimension of approximately 1×1 mm before considering reconstruction details.

Increasing matrix at the same FOV reduces pixel size and can improve spatial resolution, but the smaller voxel contains less signal. For a 2D acquisition, voxel dimensions are approximately $\text{pixel}_R \times \text{pixel}_P \times \text{slice thickness}$. Thus, spatial-resolution choices interact directly with SNR. Increasing the phase matrix also commonly increases scan time because more phase-encoding measurements are required.

Why frequency and phase resolution may differ

A radiographer can prescribe a rectangular FOV or asymmetric matrix. The phase and frequency pixel dimensions therefore do not have to be identical. Understanding which axis is phase and which is frequency helps you predict which parameter change affects scan time, aliasing risk and artifact direction.

12 Why radiographers swap phase and frequency directions

The phase and frequency axes are not merely theoretical labels. Changing them can move artifacts away from important anatomy or change the practical balance between FOV, scan time and distortion.

12.1 Phase wrap / aliasing

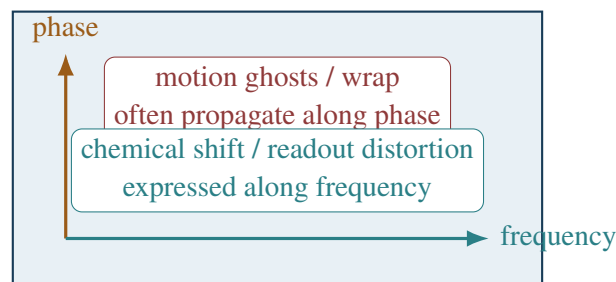
If anatomy extends beyond the encoded FOV in the phase direction, the available phase sampling cannot uniquely represent those outside positions and their signal may be mapped to the opposite side of the image, producing wrap-around or aliasing. Phase wrap is common because the phase FOV is often reduced to shorten acquisition or improve efficiency.

12.2 Motion ghosting

Repeated phase-encoding measurements are acquired at different times. If anatomy changes position or signal intensity between those measurements, the inconsistency is interpreted as spatial information and can produce replicated ghosts predominantly along the phase direction in many conventional Cartesian sequences. Respiratory motion, vascular pulsation and patient movement are common examples.

12.3 Chemical shift and readout distortion

Chemical shift displacement is classically expressed along the frequency-encoding direction because fat and water resonate at slightly different frequencies. Susceptibility-related frequency offsets can also produce geometric distortion along the readout direction, particularly when receiver bandwidth is low.



Practical planning

If motion ghosting would cross the anatomy of interest, changing the phase direction may redirect the ghosts. If wrap risk is high, the radiographer may increase phase FOV, use phase oversampling/no-wrap options, reposition the patient, add saturation strategies or select another encoding direction depending on the sequence and anatomy.

13 From encoding to k-space

Frequency and phase encoding do not directly paint pixels onto the screen. The sampled MR signal is stored as raw spatial-frequency data in **k-space**. In conventional Cartesian imaging, one readout acquires one line while a chosen phase-encoding amplitude determines which line is sampled; repeating the sequence with new phase amplitudes fills additional lines. Session 4 develops this process in detail.

Key Takeaways

- Frequency/readout encoding is active while the signal is sampled and makes frequency depend on position along one in-plane axis.
- A temporary phase gradient leaves position-dependent phase offsets; repeated phase amplitudes resolve the second in-plane axis and strongly influence scan time.
- Receiver bandwidth is a readout parameter: increasing it generally reduces frequency-related displacement/distortion but lowers SNR.
- Encoding direction matters clinically: wrap and many motion ghosts occur along phase, whereas chemical-shift/readout displacement occurs along frequency.

Key Terms

Frequency/readout encoding

Spatial encoding that makes frequency depend on position while the signal is sampled.

Phase encoding

Spatial encoding that creates position-dependent phase offsets before readout.

Gradient area

Product/integral of gradient amplitude over time; determines accumulated phase for phase encoding.

Receiver bandwidth

Frequency range accepted during MR signal readout.

Dwell time

Time between successive readout samples; inversely related to sampling bandwidth.

Phase wrap

Aliasing caused when anatomy lies outside the encoded phase FOV.

Motion ghosting

Repeated displacement of signal, commonly along phase, due to inconsistency between phase-encoded measurements.

Chemical shift

Frequency difference between fat and water that can cause readout-direction spatial displacement.

Self-Assessment Questions

1. Why is slice selection alone insufficient for a 2D image?
2. How does the readout gradient make frequency depend on position, and why can it not distinguish rows as well as columns?
3. How does a temporary phase gradient leave a lasting spatial phase pattern?
4. Why are many phase-encoding amplitudes required in conventional Cartesian MRI?
5. Distinguish transmit bandwidth from receiver bandwidth.
6. Which encoding direction would you consider changing if motion ghosts cross the anatomy of interest, and why?

Series navigation

Next session: K-Space Filling. We will use the repeated phase-encoding measurements and readout samples from this session to explain how MRI fills raw data space before Fourier reconstruction.

